

CS 171: Discussion Section 4 (9/29)

1 Pseudorandom Functions

Let $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a pseudorandom function. Prove whether the candidate following candidate is pseudorandom or not:

$$f'_{(k_1, k_2)}(x) = f_{k_1}(x) \parallel f_{k_2}(x).$$

2 CPA-Secure Encryption

Let $(\text{Gen}, \text{Enc}, \text{Dec})$ be a CPA-secure encryption scheme. Below, we will construct another encryption scheme and prove that it is also CPA-secure.

In the encryption scheme below, let the message m belong to $\{0, 1\}^n$.

- $\text{Gen}_1(1^n)$: Sample the key as follows: $k \leftarrow \text{Gen}(1^n)$.
- $\text{Enc}_1(k, m)$: Sample $r \leftarrow \{0, 1\}^n$ uniformly at random. Then compute $c_0 := \text{Enc}(k, r)$ and $c_1 := r \oplus m$. Output the ciphertext $c = (c_0, c_1)$.
- $\text{Dec}_1(k, (c_0, c_1))$: **Unspecified**

1. Fill in the decryption algorithm so that every ciphertext is decrypted correctly.

$\text{Dec}_1(k, (c_0, c_1))$:

2. Prove that $(\text{Gen}_1, \text{Enc}_1, \text{Dec}_1)$ satisfies CPA security.